Toward a four-dimensional Yang–Mills construction

Notebook 09: the flat fine inverse and a current neighborhood

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Abstract

We prove exponential locality of the full homogeneous flat fine inverse in the scalar gauge used by the deposited source account, including every placement of at most two fine differences. An explicit rank-one correction cancels its apparent massless poles. A separate constrained scalar inverse controls the differentiated gauge vertices. Together these estimates yield a complex neighborhood of bounded auxiliary currents with radius independent of fine refinement and periodic volume. The coarse precision has an exact quadratic remainder at zero current, factorial bounds at every order, and exponential paths through localized current supports. All operator norms use physical cell volume. Notebook 11 separately treats the nonlinear constraint-current term at flat background. Nonflat backgrounds and the interacting continuum limit remain open.

Status. Research proofs checked by separate internal derivations and audits. No external certification or novelty claim is made. Operator matching refers to the pinned deposited formulas; it does not reproduce every unavailable primary-paper argument. Throughout, complex adjoints mean the analytic continuation of the real bilinear transpose, rather than conjugation of a varying complex parameter.

1 Scope and normalization

The dimension is fixed, with $d = 4$ the intended application. Work on the infinite homogeneous lattice or a compatible periodic regulator. The physical coarse spacing is one and the fine spacing is M^{-1}, where M is odd. Fine inner products use measure M^{-d} and coarse inner products use counting measure. The normalized averages are

$$(Q_0\lambda)(n) = M^{-d} \sum_{s \in I_M^d} \lambda(n + s/M), \quad I_M = \{-(M-1)/2, \dots, (M-1)/2\},$$

$$(Q_M A)_\mu(n) = M^{-d-1} \sum_{s \in I_M^d} \sum_{j=0}^{M-1} A_\mu(n + (s + je_\mu)/M).$$

Write $Q = Q_M$. The physical Maxwell precision q_M is defined by

$$\langle B, q_M B \rangle = \inf_{QA=B} \|d_\eta A\|_\eta^2 .$$

Notebook 07 proves its uniform holomorphic symbol and cell locality. Notebook 08 constructs its unique source-gauge extension H_{src} and verifies the deposited averaging and gauge conventions. The following proof controls the full fine inverse required to perturb that extension. Its positive coarse penalty is an auxiliary constraint penalty: the resulting fine inverse gap is not a physical mass gap.

We use the alias estimates and physical Fourier normalization proved in Notebook 07. Relevant multiplier definitions and cancellations are recalled below. Every refinement-uniform bound is for a fixed compact positive interval of penalties, or for the source value $a = 1$.

2 The full flat source inverse

We now control the fine inverse itself, rather than only the minimized coarse action or its retained-coordinate covariance. Fine spacing is $\eta = M^{-1}$, where M is odd, and the coarse physical spacing is one. All fine pairings use volume weight η^d, while coarse pairings use unit counting measure. Let D be the fine scalar gradient, $\mathcal{L} = D^*D$ its positive scalar Laplacian, and K the fine Maxwell precision. Thus $KD = 0$ and $K + DD^* = \mathcal{L}I$. Let $Q = Q_M^{\mathrm{st}}$ be the normalized one-form straight-string average and Q_0 the centered scalar average, so that $QD = d_c Q_0$.

The scalar source gauge is

$$G' = (\mathcal{L} + Q_0^* a' Q_0)^{-1}, \qquad \mathcal{P} = G' Q_0^* (Q_0 G'^2 Q_0^*)^{-1} Q_0 G', \qquad R = I - \mathcal{P}. \tag{1}$$

The scalar projection $\mathcal{P}$ must not be confused with a Fourier denominator P_r below. The positive scalar penalty a' in this definition is distinct from the one-form penalty a in

$$\mathcal{A}_a = K + DRD^* + Q^* a Q, \qquad G_0 = \mathcal{A}_a^{-1}. \tag{2}$$

We first allow a fixed positive a, uniformly in a fixed interval $0 < a_- \le a \le a_+ < \infty$. The source convention of Raw I, lines 13583–13593, is $a = \ell^{-2}$; in the present homogeneous unit-scale normalization it is exactly $a = 1$.

Theorem 2.1. *The inverse in (2) exists on the homogeneous infinite lattice and every compatible periodic regulator. Its physical-cell kernel satisfies, for fixed positive C, ρ,*

$$\|(\mathcal{D}_\alpha G_0 \mathcal{D}_\beta)(n,m)\|_{\mathrm{cell},2\to\mathrm{cell},2} \le C e^{-\rho|n-m|_1}, \qquad \alpha + \beta \le 2. \tag{3}$$

Here n, m are unit coarse cells, each $\mathcal{D}$ is a specified composition of fine forward or backward differences of its indicated order, and fixed curl or divergence combinations are included. Constants are independent of odd M, volume, and $a \in [a_-, a_+]$. The cell Hilbert norm is the fine physical-volume norm, including the fixed orientation and color fibers. At every smaller positive exponent, (3) gives a uniform exponentially weighted cell-Schur bound. The periodic kernels are exact periodizations and obey the same bounds in periodic distance.

If q_M denotes the physical Maxwell minimum, then the penalized coarse inverse and normalized extension satisfy

$$q_0 := (QG_0Q^*)^{-1} = q_M + aI, \qquad G_0 Q^* q_0 = H_{\mathrm{src}}, \tag{4}$$

where H_{src} is the exact Maxwell minimizing extension in the matched scalar source gauge $RD^*H_{\mathrm{src}} = 0$.

The theorem is specific to the homogeneous flat source definitions. It is not a theorem for arbitrary nonflat backgrounds, Dirichlet domains, or mixed retained histories. Fine derivative bounds in (3) are cell L^2 operator bounds, not pointwise regularity assertions.

2.1 Exact aliases and the penalized Feynman inverse

Let $I_M = \{-(M-1)/2, \dots, (M-1)/2\}$, $p_r = z + 2\pi r$, and $r \in I_M^d$. Use

$$u_{r,\mu}^{\pm} = M(e^{\pm i p_{r,\mu}/M} - 1), \quad P_r = \sum_\mu u_{r,\mu}^+ u_{r,\mu}^-, \quad d_\mu^{\pm} = e^{\pm i z_\mu} - 1, \quad b_\mu = d_\mu^+ d_\mu^-.$$

More explicitly, the averaging multipliers are

$$a_r = \prod_{\nu=1}^{d} \left(M^{-1} \sum_{s \in I_M} e^{i p_{r,\nu} s/M} \right), \qquad \omega_{r,\mu} = a_r M^{-1} \sum_{j=0}^{M-1} e^{i p_{r,\mu} j/M}.$$

Write $a_r = a_M(p_r)$ for the centered scalar averaging multiplier and $\omega_{r,\mu}$ for the one-form multiplier, including its additional longitudinal string factor. Superscript minus means evaluation at the negative full fine momentum; it agrees with complex conjugation on real momenta only. String telescoping gives

$$\omega_{r,\mu} u^+_{r,\mu} = a_r d^+_\mu, \qquad \omega^-_{r,\mu} u^-_{r,\mu} = a^-_r d^-_\mu. \tag{5}$$

Alias space has counting Hilbert norm. With physical-volume Parseval normalization, no additional power of M occurs in an alias covariance sum.

Work initially in a real nonzero coarse fiber. The scalar source projection is the orthogonal projection onto $\mathrm{ran}(\mathcal{L}^{-1}Q_0^*)$. Therefore

$$\mathcal{P} = \mathcal{L}^{-1} Q_0^* W_2^{-1} Q_0 \mathcal{L}^{-1}, \qquad W_2 = \sum_r \frac{a_r a^-_r}{P_r^2}. \tag{6}$$

This agrees with (1): multiplying by the positive scalar penalty changes the coarse parametrization of $\mathrm{ran}(G'Q_0^*)$, but not that range. Its constant scalar mode will be treated through the original penalized definition.

Define

$$G_F = \mathrm{diag}_r(P_r^{-1} I), \qquad W = Q G_F Q^* = \mathrm{diag}_\mu(c_\mu), \qquad c_\mu = g_\mu / P_0 + R_\mu,$$

$$g_\mu = \omega_{0,\mu} \omega^-_{0,\mu}, \qquad R_\mu = \sum_{r \neq 0} \frac{\omega_{r,\mu} \omega^-_{r,\mu}}{P_r}.$$

The Feynman operator with the one-form penalty is $\mathcal{B} = \mathcal{L} I + Q^* a Q$. Its inverse obeys

$$F := \mathcal{B}^{-1} = G_F - G_F Q^* (a^{-1} I + W)^{-1} Q G_F. \tag{7}$$

Put

$$\widehat{R}_\mu = R_\mu + a^{-1}, \qquad \mathcal{D}_\mu = g_\mu + P_0 \widehat{R}_\mu, \qquad k_\mu = \mathcal{D}_\mu^{-1}. \tag{8}$$

Then $(a^{-1} I + W)^{-1} = P_0 \,\mathrm{diag}(k_\mu)$ and

$$F_{r\mu,s\nu} = \delta_{\mu\nu} \left[\frac{\delta_{rs}}{P_r} - \frac{P_0 \omega^-_{r,\mu} \omega_{s,\mu}}{P_r P_s \mathcal{D}_\mu} \right]. \tag{9}$$

The central blocks cancel exactly to

$$\begin{aligned} F_{0\mu,0\nu} &= \delta_{\mu\nu} \widehat{R}_\mu / \mathcal{D}_\mu, \\ F_{0\mu,s\nu} &= -\delta_{\mu\nu} \frac{\omega^-_{0,\mu} \omega_{s,\mu}}{P_s \mathcal{D}_\mu} \qquad (s \neq 0), \\ F_{r\mu,0\nu} &= -\delta_{\mu\nu} \frac{\omega^-_{r,\mu} \omega_{0,\mu}}{P_r \mathcal{D}_\mu} \qquad (r \neq 0). \end{aligned} \tag{10}$$

Use (9) only with both aliases high, and (10) otherwise. These expressions contain no central massless denominator.

2.2 The scalar projection and the second Woodbury identity

The Hodge identity and $R = I - \mathcal{P}$ give

$$\mathcal{A}_a = \mathcal{B} - vW_2^{-1}v^-, \qquad v_{r,\mu} = u^+_{r,\mu}a^-_r/P_r, \quad v^-_{r,\mu} = a_r u^-_{r,\mu}/P_r. \tag{11}$$

Here v^- is a row, the analytic transpose at negative arguments. On real fibers it is the Hilbert adjoint of v.

The real fine precision is positive definite. Indeed, a vector in its kernel has zero curl, zero coarse average, and zero source-gauge term. The zero-curl field is $D\lambda + c$, with c a constant harmonic one-form. Taking the coarse mean of $Q(D\lambda + c) = 0$ gives $c = 0$; subtracting a scalar constant makes $Q_0\lambda = 0$. Then $\mathcal{L}\lambda \in \operatorname{ran} R$, while the source-gauge condition says $R\mathcal{L}\lambda = 0$. Thus $D\lambda = 0$. This proves finite-regulator positivity with the harmonic sector retained. The subtractive Woodbury identity therefore gives

$$G_0 = F + Fv\,(W_2 - v^-Fv)^{-1}v^-F. \tag{12}$$

Equation (5) implies the three exact contractions

$$v^-G_Fv = W_2, \qquad QG_Fv = W_2d^+, \qquad v^-G_FQ^* = W_2d^-. \tag{13}$$

For example, the μ component of the middle term is $\sum_r \omega_{r,\mu}u^+_{r,\mu}a^-_r/P_r^2 = W_2d^+_\mu$. Let $\mathfrak{a} = a_0a_0^-$. The central cochain identity is $\sum_\mu b_\mu/g_\mu = P_0/\mathfrak{a}$. Define

$$\mathcal{E}_a = \mathfrak{a}^{-1} - \sum_\mu \frac{b_\mu \widehat{R}_\mu}{g_\mu \mathcal{D}_\mu}. \tag{14}$$

Subtracting $1/\mathcal{D}_\mu - 1/g_\mu$ proves

$$\sum_\mu b_\mu k_\mu = P_0\mathcal{E}_a. \tag{15}$$

It follows that

$$W_2 - v^-Fv = W_2^2d^-(a^{-1}I + W)^{-1}d^+ = W_2^2P_0^2\mathcal{E}_a. \tag{16}$$

Introduce the scalar second-order remainder and its canceled denominator

$$R_{s,2} = \sum_{r\neq 0}\frac{a_ra^-_r}{P_r^2}, \qquad D_{s,2} = \mathfrak{a} + P_0^2R_{s,2}, \qquad W_2 = D_{s,2}/P_0^2. \tag{17}$$

Thus (16) is $D_{s,2}^2\mathcal{E}_a/P_0^2$. In particular it contains W_2^2, not W_2.

2.3 The final canceled inverse

Set $\beta = (P_0/D_{s,2})Fv$. For a high alias,

$$\beta_{r,\mu} = \frac{P_0u^+_{r,\mu}a^-_r}{P_r^2D_{s,2}} - \frac{\omega^-_{r,\mu}k_\mu d^+_\mu}{P_r}, \qquad r \neq 0. \tag{18}$$

For the central entry use

$$\omega^-_{0,\mu}k_\mu d^+_\mu = (u^+_{0,\mu}/a_0)g_\mu k_\mu,$$

$$\mathfrak{a}/D_{s,2} = 1 - P_0^2R_{s,2}/D_{s,2}, \qquad g_\mu k_\mu = 1 - P_0\widehat{R}_\mu k_\mu.$$

The resulting exact central formula is

$$\beta_{0,\mu} = \frac{u^+_{0,\mu}}{a_0}\left(\widehat{R}_\mu k_\mu - \frac{P_0 R_{s,2}}{D_{s,2}}\right). \tag{19}$$

The row $\beta^- = (P_0/D_{s,2})v^- F$ is its analytic transpose: in the high formula it has entries $P_0 a_r u^-_{r,\mu}/(P_r^2 D_{s,2}) - d^-_\mu k_\mu \omega_{r,\mu}/P_r$; centrally use $u^-_{0,\mu}/a^-_0$ times the same bracket. Equations (12) and (16) now become

$$\boxed{G_0 = F + \beta \mathcal{E}_a^{-1} \beta^-.} \tag{20}$$

Every central pole has canceled. The update has a positive sign, and the second term in the bracket of (19) has a negative sign. The formula removes the entire complex cone $P_0 = 0$.

2.4 A strip and constants uniform in refinement

Choose $\pi < K_0 < 2\pi$ and set

$$c_* = \frac{\sin(K_0/2)}{K_0/2}, \qquad H_* = \frac{4(2\pi - K_0)^2}{\pi^2}.$$

On the enlarged real cube $|x_\mu| \le K_0$, the alias estimates of Notebook 07 give

$$\mathfrak{a} \ge c_*^{2d}, \quad g_\mu \ge g_* := c_*^{2d+2}, \quad 0 \le R_\mu \le H_*^{-1}, \quad 0 \le R_{s,2} \le H_*^{-2}, \quad P_0 \le dK_0^2,$$

with $g_\mu, \mathfrak{a} \le 1$. Consequently

$$g_* \le \mathcal{D}_\mu \le D_* := 1 + dK_0^2(H_*^{-1} + a_-^{-1}), \qquad c_*^{2d} \le D_{s,2} \le 1 + d^2 K_0^4 H_*^{-2}.$$

For $x \neq 0$, (15) gives

$$\mathcal{E}_a(x) = \frac{\sum_\mu b_\mu(x)/\mathcal{D}_\mu(x)}{P_0(x)} \ge c_*^2/D_*. \tag{21}$$

Here $\sum b_\mu \ge c_*^2 |x|^2$ and $P_0 \le |x|^2$. At zero the defining expression gives $\mathcal{E}_a(0) = 1$.

On an initial fixed complex neighborhood, high denominators obey $|P_r(z)| \ge c|r|^2$ for $r \neq 0$, including the edge aliases. Finite alias Parseval bounds the sums of absolute paired scalar and one-form averaging factors independently of M. Hence R_μ, $R_{s,2}$, and the central multipliers are uniformly holomorphic and bounded. The added a^{-1} is bounded by the hypothesis on a.

Use nested real cubes with positive margins. Uniform Cauchy estimates first allow an imaginary shrinking that keeps $\mathfrak{a}, g_\mu, \mathcal{D}_\mu, D_{s,2}$ separated from zero. The expression (14) is then holomorphic and uniformly bounded. A second shrinking, using (21), keeps $\mathcal{E}_a$ separated from zero. This gives one positive strip width independent of M, a, with the real fundamental cube in the interior of each enlarged chart. The inverse of a_0 is also bounded because $a_0 a_0^- = \mathfrak{a}$ is separated from zero and both factors are bounded. Real positivity has thus been combined with uniform analytic estimates; it is not being used alone to claim complex invertibility.

2.5 Zero mode and the independent $M = 1$ check

At coarse momentum zero, $a_0 = \omega_{0,\mu} = 1$, all high $a_r, \omega_{r,\mu}$ vanish, and

$$R_\mu = R_{s,2} = 0, \qquad \mathfrak{a} = g_\mu = \mathcal{D}_\mu = D_{s,2} = \mathcal{E}_a = 1, \qquad \beta = 0.$$

The canceled formula gives

$$(G_0)_{0\mu,0\nu} = a^{-1}\delta_{\mu\nu}, \qquad (G_0)_{r\mu,s\nu} = \delta_{rs}\delta_{\mu\nu}/P_r \quad (r, s \neq 0), \tag{22}$$

with zero mixed blocks. The source projection R is zero on the constant scalar mode and identity on high scalar aliases. Hence the central precision is aI, and the high precision is $P_r I$, exactly as in (22). No scalar constant Laplacian was inverted.

For $M = 1$, $Q = I$, $R = 0$, and

$$\mathcal{D} = 1 + P_0/a, \quad k = \mathcal{E}_a = a/(P_0 + a), \quad F = (P_0 + a)^{-1} I, \quad \beta = d^+/(P_0 + a).$$

Thus

$$G_0 = (P_0 + a)^{-1} I + \frac{d^+ d^-}{a(P_0 + a)}, \tag{23}$$

the direct inverse of $(P_0 + a)I - d^+ d^-$. This checks the sign, penalty normalization, and rank-one coefficient. The value $a^{-1} I$ at zero also shows why a uniform lower bound for a is essential.

2.6 Alias bounds with two fine derivatives

On the common strip,

$$|P_r| \geq c|r|^2 \ (r \neq 0), \quad |u^{\pm}_{r,\mu}| \leq C(1 + |r|), \quad \sum_r |a^{\pm}_r|^2 + \sum_{r,\mu} |\omega^{\pm}_{r,\mu}|^2 \leq C_d.$$

The high-high block of F is a bounded diagonal operator plus one rank-one term for each orientation. Its high rank vectors are $\omega^{\pm}_{r,\mu}/P_r$, and its mixed row and column use the same vectors. Their alias L^2 norms are bounded. The central block is bounded separately. Moreover (18) gives

$$|\beta_{r,\mu}| \leq C \left(\frac{|a^-_r|}{|r|^3} + \frac{|\omega^-_{r,\mu}|}{|r|^2} \right), \qquad r \neq 0. \tag{24}$$

The central beta is bounded by (19). Therefore F and G_0 have uniformly bounded alias Hilbert operator norms. The norm estimate for an analytic rank-one matrix vw^T uses $\|v\|_2 \|w\|_2$, without conjugating its analytic parameters.

With α differences on the output and β on the input, $\alpha + \beta \leq 2$, the high diagonal of F grows at most as $|r|^{\alpha+\beta-2}$, which is bounded. Each rank vector has at most two extra powers of $|r|$ multiplying $\omega^{\pm}_{r,\mu}/P_r$, so its L^2 norm is still bounded by Parseval. The mixed blocks have the same property. For the update in (20), two derivatives on a beta column are bounded, in the worst case, by $C(|a^-_r|/|r| + |\omega^-_{r,\mu}|)$. This is square summable uniformly, as is the corresponding row. Every derivative sandwich of total order at most two is consequently holomorphic and uniformly bounded in alias operator norm. The number of derivative-orientation components is fixed in d.

2.7 Reconstruction, periodic gluing, and the actual inverse

For within-cell sites $s \in I_M^d$ use $\mathcal{F}_M(z)_{sr} = e^{i(z+2\pi r)\cdot s/M}$. For real z it is unitary from alias counting L^2 into the cell norm $M^{-d} \sum_s |v_s|^2$. For complex $z = x + iy$, it differs from that unitary by $\mathrm{diag}_s(e^{-y\cdot s/M})$. Its norm and the norm of its holomorphic inverse are thus uniformly bounded. Tensoring with the orientation identity, the reconstructed symbol is

$$(G_0)_{\mathrm{cell}}(z) = \mathcal{F}_M(z) G_0(z) \mathcal{F}_M(z)^{-1}.$$

All derivative sandwiches inherit its uniform strip bounds.

The full real inverse, rather than each separated central term, is periodic up to alias permutation. Reconstruction cancels the permutation; wrapping an alias gives $e^{i2\pi s_\nu} = 1$. The reconstructed symbol is therefore periodic. The enlarged-chart holomorphic formulas agree on real overlaps and glue by

the identity theorem. Fourier contour shifting proves (3). Summing at a smaller exponent proves the weighted Schur bounds. These are operator-valued contour estimates, so no factor growing with the within-cell dimension is introduced.

For each fixed M, fiberwise multiplication, with the zero sector as in (22), verifies that this bounded kernel is the inverse of (2) on the infinite homogeneous Hilbert space. The real inverse bound yields a uniform positive fine spectral gap. Sampling the same symbol on a compatible periodic momentum grid gives exactly the finite source inverse. Absolute summability identifies its kernel with the periodization; periodic distance bounds its weighted sums by the infinite sums. This includes short periods with their proper oriented-cell multiplicities.

Finally, at fixed coarse datum B, the Maxwell energy is at least $\langle B, q_M B\rangle$, the source gauge term is nonnegative, and the penalty is the constant $a\,\|B\|^2$. The source-gauge minimizer $H_{\mathrm{src}}B$ attains both minima. Therefore

$$\inf_{QA=B}\langle A, \mathcal{A}_a A\rangle = \langle B, (q_M + aI)B\rangle.$$

The constrained Gaussian inverse formula proves $q_0 = q_M + aI$ and, by strict positivity and uniqueness, $G_0 Q^* q_0 = H_{\mathrm{src}}$. In the source convention $QH_0 = \ell^{-1}$, the normalized extension is $H_0\ell$. This proves (4) and completes the theorem.

Remark 2.2. The homogeneous source passage fixes $a = \ell^{-2}$ and hence $a = 1$ here. Raw I, lines 23879–23881, separately introduces an a_n sequence and refers to CMP99b (3.23)–(3.24) for an unprinted recursion. The theorem does not guess that sequence: an application using it must verify the fixed positive interval after its measure and scale conversion. The scalar penalty a' in the projection remains a different object. General nonflat and retained-history identifications are not consequences of the homogeneous Fourier proof.

3 The scalar gauge inverse and bounded current vertices

The fine inverse alone does not bound the current perturbation. The source gauge contains derivatives, so we first control its scalar constrained inverse in the same physical cell norms. Keep the matched scalar average Q_0, scalar Laplacian $\mathcal{L} = D^*D$, and the source operators

$$G' = (\mathcal{L} + Q_0^* a' Q_0)^{-1}, \qquad R = I - G'Q_0^*(Q_0 G'^2 Q_0^*)^{-1} Q_0 G', \qquad Z = G'RG'.$$

All adjoints use the physical measures. Complex continuations use the fixed bilinear transpose, never conjugation of a varying parameter.

3.1 The constrained biharmonic inverse

The operator

$$Z = G'^2 - G'^2 Q_0^*(Q_0 G'^2 Q_0^*)^{-1} Q_0 G'^2$$

is the inverse of the quadratic form $\|\mathcal{L}\lambda\|^2$ on $\ker Q_0$, extended by zero on its orthogonal complement. Indeed it is the usual constrained inverse of G'^{-2}, and for $Q_0\lambda = 0$ one has $\|G'^{-1}\lambda\|^2 = \|\mathcal{L}\lambda\|^2$. It follows that $Q_0 Z = 0 = ZQ_0^*$ and that Z is independent of the auxiliary positive penalty a'. The constant scalar mode is excluded by the constraint, so no inverse is assigned to that mode.

On a real nonzero Bloch fiber the constrained inverse is

$$Z_{rs} = \frac{\delta_{rs}}{P_r^2} - \frac{a_r^- a_s}{P_r^2 P_s^2 W_2}, \qquad W_2 = \frac{D_{s,2}}{P_0^2}.$$

Its completely canceled form is

$$Z_{00} = \frac{R_{s,2}}{D_{s,2}}, \qquad Z_{0s} = -\frac{a_0^- a_s}{P_s^2 D_{s,2}} \quad (s \neq 0), \tag{25}$$

$$Z_{r0} = -\frac{a_r^- a_0}{P_r^2 D_{s,2}} \quad (r \neq 0), \qquad Z_{rs} = \frac{\delta_{rs}}{P_r^2} - \frac{P_0^2 a_r^- a_s}{P_r^2 P_s^2 D_{s,2}} \quad (r, s \neq 0). \tag{26}$$

The notation $D_{s,2}$ denotes the scalar denominator, not a dependence on the second matrix index s. No central inverse power of P_0 remains. At zero coarse momentum the high averaging coefficients vanish: Z is zero on the constant mode and equals P_r^{-2} on the nonzero aliases. For $M = 1$, it is identically zero.

Proposition 3.1. *Every placement of at most four total fine forward or backward differences on the two sides of Z has a uniformly bounded holomorphic physical-cell symbol and a uniformly exponentially decaying cell-operator kernel. The bounds are independent of refinement and compatible periodic volume.*

Proof. Let the output order be j and the input order be l, with $j + l \leq 4$. The high diagonal in (26) is bounded by $C|r|^{j+l-4}$. Its high rank vectors are bounded by $C|a_r^-||r|^{j-4}$ and $C|a_r||r|^{l-4}$. Each order is at most four, so alias Parseval bounds both vector Hilbert norms uniformly. The central row and column use the same vectors; central multipliers are bounded. The scalar denominator $D_{s,2}$ stays uniformly away from zero on the common strip. For a complex rank-one matrix vw^T, its Hilbert norm is bounded by $\|v\|_2 \|w\|_2$, without conjugating the analytic symbol.

These are operator bounds before reconstruction and hence do not count the growing number of aliases. The finite Fourier reconstruction and its inverse are uniformly bounded in physical cell Hilbert norm, with the inverse containing its required M^{-d} factor. Alias permutation and the integer-site phases give periodic gluing on the enlarged charts. Contour shifting gives exponential cell decay; summing at a smaller exponent gives weighted Schur bounds. Fourier sampling gives exact periodizations on finite tori, including the stated zero mode and small-period incidence multiplicities. □

The source definitions give

$$Q_0 G' R = 0, \quad \mathcal{L} G' R = R, \quad R G' \mathcal{L} = R.$$

Consequently

$$\Lambda := G' R D^* = Z \mathcal{L} D^*, \qquad D\Lambda = D Z \mathcal{L} D^*.$$

The total fine derivative orders are three and four, respectively. Proposition 3.1 therefore bounds both operators in a common positive weighted cell-Schur norm. This is a Hilbert operator conclusion; it does not assert pointwise bounds for four fine derivatives.

4 A uniform neighborhood for the flat auxiliary current

We keep the background equal to the identity and vary the current independently. Fine spacing is M^{-1}, with odd M, and the fine measure is M^{-d}. The one-form average is the corrected source average at flatness, already identified with $Q = Q_M^{\mathrm{st}}$. The scalar source projection is R, and D is the fine scalar gradient. Write

$$\Delta_0 = d_\eta^* d_\eta + D R D^* + Q^* a Q, \qquad G_0 = \Delta_0^{-1}, \qquad q_0 = (Q G_0 Q^*)^{-1}.$$

The homogeneous source convention has $a = 1$ at unit physical block scale. All conclusions below also hold uniformly for a fixed compact interval $0 < a_- \leq a \leq a_+ < \infty$. We do not assert a bound for an unmatched sequence of penalties.

The preceding flat inverse theorem and minimizing-extension theorem give

$$q_0 = q_M + aI, \qquad E_0 = G_0 Q^* q_0, \qquad QE_0 = I, \qquad RD^* E_0 = 0. \tag{27}$$

Here q_M is the unpenalized constrained Maxwell precision. The addition of aI must be retained in q_0. The normalized extension E_0 is unchanged by this penalty; a source convention $QH_0 = \ell^{-1}$ instead uses $H_0 = E_0 \ell^{-1}$.

All adjoints initially use these physical measures at the real flat reference. At complex current they denote the fixed bilinear transpose. They do not conjugate the varying complex current.

4.1 Current vertices and the weighted algebra

For a fine bond b, the flat gauge expansion has mixed and polarized quadratic terms

$$c^{(1)}(A, \lambda)(b) = \frac{i}{2}[\lambda(b_-) + \lambda(b_+), A(b)], \tag{28}$$

$$2c^{(2)}(\lambda_1, \lambda_2)(b) = \frac{1}{2i}([\lambda_1(b_-), (D\lambda_2)(b)] + [\lambda_2(b_-), (D\lambda_1)(b)]). \tag{29}$$

These are the deposited formulas at Raw I:13698–13718 and I:13811–13821. Define the current-linear maps by

$$\langle \lambda, \mathfrak{j}_J A \rangle = \langle c^{(1)}(A, \lambda), J \rangle, \qquad \mathfrak{k}_J = \mathfrak{j}_J^*,$$
$$\langle \lambda_1, \mathfrak{m}_J \lambda_2 \rangle = 2\langle c^{(2)}(\lambda_1, \lambda_2), J \rangle.$$

All bond and site pairings carry the same fine volume weight. Set $\Lambda = G' R D^*$, with the fixed scalar source inverse G'. The scalar locality lemma gives uniform weighted bounds for Λ and $D\Lambda$. The auxiliary perturbation is

$$\Theta'_J = \mathfrak{k}_J \Lambda + \Lambda^* \mathfrak{j}_J + \Lambda^* \mathfrak{m}_J \Lambda. \tag{30}$$

This is precisely the auxiliary perturbation in Raw I:13725–13731. It is distinct from the additional nonlinear-constraint operator denoted there by Θ_J.

Group fine fields into unit coarse cells with their physical Hilbert norms. For block operators between these fine fibers, or between fine and fixed-dimensional coarse fibers, use

$$\|T\|_\kappa = \max\left\{ \sup_x \sum_y e^{\kappa d(x,y)} \|T_{xy}\|, \sup_y \sum_x e^{\kappa d(x,y)} \|T_{xy}\| \right\}. \tag{31}$$

Here $d(x, y) = |x - y|_1$ in infinite volume and is periodic coarse-cell distance on a finite compatible torus. This norm is submultiplicative, including rectangular products, and transpose preserves it.

Lemma 4.1 (A current bound without derivatives of the current)**.** *There is $\theta < \infty$, independent of refinement and compatible periodic volume, such that*

$$\|\Theta'_J\|_\kappa \le \theta \|J\|_\infty. \tag{32}$$

The map $J \mapsto \Theta'_J$ is complex linear. The current requires no regularity beyond boundedness.

Proof. Define the finite-incidence map V_J from fine bonds to fine sites by

$$\langle \lambda, V_J A \rangle = \frac{1}{2i}\langle [\lambda(b_-), A(b)], J(b) \rangle.$$

Equation (29) gives the exact form identity

$$\mathfrak{m}_J = V_J D + D^* V_J^*.$$

Thus (30) becomes

$$\Theta'_J = \mathfrak{j}_J^* \Lambda + \Lambda^* \mathfrak{j}_J + \Lambda^* V_J D\Lambda + (D\Lambda)^* V_J^* \Lambda. \tag{33}$$

If $|[X,Y]| \le c_{\mathrm{br}}|X||Y|$, both $\mathfrak{j}_J$ and V_J have fine Hilbert operator norm at most $C_d c_{\mathrm{br}} \|J\|_\infty$. Indeed there are at most $2d$ incident bonds per site and two endpoints per bond; Cauchy–Schwarz and the common physical volume weight give this bound without a power of M. These coefficient maps have a fixed finite cell range, so the same estimate holds in (31).

Apply the weighted algebra inequality to (33), using the established bounds for $\Lambda, D\Lambda$. Every derivative is on a scalar leg through $D\Lambda$. No derivative acts on J, even if J jumps at a cell boundary. The proof also applies to complex currents with the fixed bilinear transpose. □

Choose a common sufficiently small $\kappa > 0$. The flat inverse, averaging and Maxwell locality results give constants g, b, q with

$$\|G_0\|_\kappa \le g, \qquad \|Q\|_\kappa, \|Q^*\|_\kappa \le b, \qquad \|q_0\|_\kappa \le q. \tag{34}$$

The fine inverse result also gives all physical fine-difference sandwiches of total order at most two. These are established homogeneous flat inputs; they are not hypotheses about a nonflat source carrier. None of these constants includes a factor for the number of sites in a fine cell.

4.2 The scalar identity behind the zero-current response

The source gauge of $E_0 = H_{\mathrm{src}}$ gives $\Lambda E_0 = 0$. For $B_0 = G_0 - E_0 q_0^{-1} E_0^*$ one has

$$G_0 \Lambda^* = B_0 \Lambda^* = DZ, \qquad \Lambda D Z = Z. \tag{35}$$

Indeed, for $\lambda = Zf$ the scalar identities give $Q_0 \lambda = 0$ and $\mathcal{L}\lambda = RG'f$, so

$$\Delta_0 D\lambda = DRG'f = \Lambda^* f, \qquad QD\lambda = 0.$$

The first equality follows by the full inverse; the constrained subtraction vanishes because $E_0^* \Lambda^* = 0$. Also $\Lambda DZ = G'R\mathcal{L}Z = G'RG' = Z$. These exact identities will identify the first extension response and the second coarse-precision response below.

4.3 Uniform inverses and the exact quadratic remainder

Define

$$c = gb^2 q, \qquad e = gbq, \qquad t_0 = g(1+c), \qquad \varepsilon = \frac{1}{4g\theta(1+c)}. \tag{36}$$

If $\theta = 0$ the perturbation vanishes and the current restriction is unnecessary. Otherwise use the displayed positive radius.

Theorem 4.2 (Uniform flat auxiliary-current neighborhood)**.** *For every complex bounded current with $\|J\|_\infty < \varepsilon$, the operators*

$$G_J = (\Delta_0 - \Theta'_J)^{-1}, \qquad q_J = (QG_J Q^*)^{-1}, \qquad E_J = G_J Q^* q_J \tag{37}$$

exist and depend holomorphically on J in the weighted algebra. They are the deposited auxiliary family, with $G_J = G_\pi^J$, $q_J = \mathsf{q}_J$ and $E_J = H^J \ell$. Uniformly in odd M, compatible volume, and $a \in [a_-, a_+]$,

$$\|G_J\|_\kappa \le 2g, \qquad \|q_J\|_\kappa \le 2q, \qquad QE_J = I, \qquad \|E_J\|_\kappa \le 4e. \tag{38}$$

Let

$$B_0 = G_0 - E_0 Q G_0 = G_0 - E_0 q_0^{-1} E_0^*. \tag{39}$$

Then the exact formulas are

$$E_J - E_0 = (I - B_0\Theta'_J)^{-1} B_0 \Theta'_J E_0, \tag{40}$$

$$q_J - q_0 = -E_0^* \Theta'_J (I - B_0\Theta'_J)^{-1} B_0 \Theta'_J E_0. \tag{41}$$

In particular,

$$\|E_J - E_0\|_\kappa \le 2t_0\theta e \, \|J\|_\infty , \qquad \|q_J - q_0\|_\kappa \le 2e^2\theta^2 t_0 \, \|J\|_\infty^2 . \tag{42}$$

Proof. Abbreviate $\Theta = \Theta'_J$. The first Neumann series gives

$$G_J = (I - G_0\Theta)^{-1} G_0, \qquad \|G_J - G_0\|_\kappa \le 2g^2\theta \, \|J\|_\infty ,$$

because $\|G_0\Theta\|_\kappa < 1/4$. Writing $W_J = QG_JQ^*$, we obtain

$$\|q_0(W_J - W_0)\|_\kappa \le 2qb^2g^2\theta \, \|J\|_\infty < \frac{c}{2(1+c)} < \frac{1}{2}.$$

The second Neumann series, with its specified order, is

$$q_J = [I + q_0(W_J - W_0)]^{-1} q_0.$$

This proves (38). Both series converge in the same Banach algebra and therefore prove holomorphy on the entire stated complex ball.

The constrained covariance has

$$QB_0 = B_0Q^* = 0, \qquad B_0\Delta_0 = I - E_0Q, \qquad \|B_0\|_\kappa \le t_0.$$

Apply B_0 to $(\Delta_0 - \Theta)E_J = Q^*q_J$ and use $QE_J = I$. This gives $E_J - E_0 = B_0\Theta E_J$, hence (40). No commutation of B_0 with Θ is used. The chosen radius has $\|B_0\Theta\|_\kappa < 1/4$.

Multiplying the stationary equation by E_0^* gives $q_J = q_0 - E_0^*\Theta E_J$. The flat source gauge implies $\Lambda E_0 = 0$, and (30) then gives

$$\Theta'_J E_0 = \Lambda^* \mathfrak{j}_J E_0, \qquad E_0^* \Theta'_J E_0 = 0. \tag{43}$$

Substitution of (40) proves (41). Finally $\|E_0\|_\kappa \le e$ and $\|(I - B_0\Theta)^{-1}\|_\kappa \le 2$ give (42). □

4.4 Factorial derivatives and localized current legs

For a current in this ball put

$$B_J = G_J - E_JQG_J, \qquad t = 2g(1 + 4c). \tag{44}$$

Then B_J is the symmetric constrained covariance and $\|B_J\|_\kappa \le t$. All directional derivatives below are derivatives in current space, not fine spatial differences.

Proposition 4.3 (Exact derivative chains)**.** *For $n \ge 1$ and bounded current directions $J_1, \dots, J_n$,*

$$D^n q_J[J_1, \dots, J_n] = -\sum_{\sigma \in S_n} E_J^* \Theta'_{J_{\sigma(1)}} B_J \Theta'_{J_{\sigma(2)}} \cdots B_J \Theta'_{J_{\sigma(n)}} E_J. \tag{45}$$

Each summand contains $n-1$ factors B_J; for $n = 1$ it is $-E_J^*\Theta'_{J_1}E_J$. The corresponding formula for D^nE_J is the sum of the ordered products $B_J\Theta'_{J_{\sigma(1)}} \cdots B_J\Theta'_{J_{\sigma(n)}}E_J$. Consequently

$$\|D^n q_J[J_1, \dots, J_n]\|_\kappa \le n!(4e)^2\theta^n t^{n-1} \prod_i \|J_i\|_\infty , \tag{46}$$

$$\|D^n E_J[J_1, \dots, J_n]\|_\kappa \le n!(4e)(t\theta)^n \prod_i \|J_i\|_\infty . \tag{47}$$

Proof. At a fixed J, repeat the stationary-equation argument for a sufficiently small increment K. It gives the exact identities

$$B_{J+K} = (I - B_J\Theta'_K)^{-1}B_J, \tag{48}$$
$$E_{J+K} = (I - B_J\Theta'_K)^{-1}E_J, \tag{49}$$
$$q_{J+K} = q_J - E_J^*\Theta'_K(I - B_J\Theta'_K)^{-1}E_J. \tag{50}$$

The first is the resolvent identity for the quadratic form restricted to $\ker Q$, with its inverse extended by zero on $\operatorname{ran} Q^*$. Expanding the series and taking the coefficient multilinear in the directions produces exactly the permutations in (45) and the extension formula. The norm bounds follow from submultiplicativity. The argument is valid at each point in the ball by using a sufficiently small increment; it does not enlarge the current domain. □

At zero current $Dq_0[J_1] = 0$ by (43). The earlier scalar gauge identity, with $Z = G'RG'$, gives the more explicit specializations

$$\begin{aligned} DE_0[J_1] &= DZ\mathfrak{j}_{J_1}E_0, \\ D^2q_0[J_1, J_2] = &- (\mathfrak{j}_{J_1}E_0)^*Z(\mathfrak{j}_{J_2}E_0) \\ &- (\mathfrak{j}_{J_2}E_0)^*Z(\mathfrak{j}_{J_1}E_0). \end{aligned} \tag{51}$$

The first extension response is a fine gradient. It need not preserve the original source gauge at nonzero current.

Proposition 4.4 (Paths through the current supports)**.** *Let J_{x_i} be supported on fine bonds based in cell x_i. There are constants C_0, C_1, independent of refinement, compatible volume, source locations and n, such that*

$$\begin{aligned} &\|D^n q_J[J_{x_1}, \dots, J_{x_n}](b, c)\| \\ &\quad \le C_0 C_1^n \prod_i \|J_{x_i}\|_\infty \sum_{\sigma \in S_n} \exp\left[-\kappa\left(d(b, x_{\sigma(1)}) + \sum_{i=1}^{n-1} d(x_{\sigma(i)}, x_{\sigma(i+1)}) + d(x_{\sigma(n)}, c)\right)\right]. \end{aligned} \tag{52}$$

Here b, c are coarse cells with their fixed orientation/color fibers. Repeated source cells are allowed. The extension derivatives have the same path structure, with the appropriate fine/coarse end fibers.

Proof. For a current supported in cell x, each coefficient vertex $\mathfrak{j}_{J_x}$ or V_{J_x} reads and writes only a fixed neighboring-cell set of x, and its restricted operator norm is $O(\|J_x\|_\infty)$. Expand each perturbation in a chain using (33). Between consecutive local vertices are products of the form AB_JC^*, with the appropriate scalar/bond domains and with A, C among $I, \Lambda, D\Lambda$. Such products have a uniform weighted Schur bound and hence their (x, y) block is bounded by a constant times $e^{-\kappa d(x,y)}$. The end products use AE_J and obey the same statement.

There are at most four choices per perturbation and a fixed number of neighboring cells per coefficient vertex. Their sizes and bounded offsets are absorbed into C_1^n. Multiplying the block bounds and then summing the permutations proves (52). This uses physical block operator norms before multiplying; it never counts the fine sites in a block. It also avoids convolution of pointwise bounds at a borderline exponential rate. Periodic distance obeys the same triangle inequality, so the proof is unchanged on compatible tori. □

4.5 Fine derivatives, finite volume, and the scope of the result

The proved full flat inverse theorem controls every specified fine difference sandwich of total order at most two. Write $\mathcal{D}_\alpha$ and $\mathcal{D}_\beta$ for fixed words of those orders, with the appropriate orientation domains. The exact identity

$$\begin{aligned}\mathcal{D}_\alpha G_J \mathcal{D}_\beta &= \mathcal{D}_\alpha G_0 \mathcal{D}_\beta \\ &\quad + (\mathcal{D}_\alpha G_0)\Theta'_J (I - G_0\Theta'_J)^{-1} G_0 \mathcal{D}_\beta\end{aligned} \tag{53}$$

therefore gives the same uniform weighted bounds when $\alpha + \beta \le 2$. Derivatives act on the outside propagator factors, not on the current. The one-sided estimates also give the corresponding output fine derivatives of E_J. These are physical cell L^2 operator estimates; no pointwise or Hölder carrier estimate is inferred from them.

The reference kernels on finite compatible tori are exact periodizations of the homogeneous kernels. All subsequent arguments take place in the weighted block algebra on that torus. They apply to arbitrary bounded periodic currents, without requiring them to be constant from cell to cell, and retain constants uniform even for short periods. The infinite-volume formulas follow in the same algebra. No passage through an unweighted growing-dimensional matrix norm is used.

The result is a uniform complex neighborhood for the displayed *auxiliary* flat family, with a quadratic precision remainder, factorial current derivatives, and paths through their current supports. The full source inverse instead has

$$G_1^J = [\Delta_0 - \Theta'_J - \Pi^* \Theta_J \Pi]^{-1}, \qquad \Pi = I - D\Lambda.$$

The separate nonlinear-constraint term Θ_J is not covered here. Its first coarse-precision derivative can already contain $-E_0^* \Theta_J^{[1]} E_0$. Thus the cancellation (43) is not a statement that the full physical precision has zero current derivative. Nonflat backgrounds, retained multiscale boundaries, source pointwise carrier norms, and the interacting ultraviolet limit remain outside this theorem.

5 What has been discharged and what is next

The flat auxiliary-current family now has a verified weighted complex domain, rather than only formal derivatives at zero. The inverse and extension bounds can supply propagator legs for subsequent estimates, and the ordered source paths identify where current derivatives are spatially supported. This discharges a specific homogeneous input to the response program in Notebook 03.

Notebook 11 now bounds the additional operator $\Pi^* \Theta_J \Pi$ in this weighted Hilbert norm along $M = L^j$. Its proof retains the corrected nonlinear one-step map and composes its quadratic jets with their physical scale factors; a different rooted one-step average is not substituted. Nonflat coefficients also change the covariant derivatives, scalar gauge, and averaging map. These changes require a matched uniform domain, not just a finite-regulator analytic inverse. Pointwise source carrier estimates and retained-history boundary estimates remain separate. Neither the free ultraviolet limit of Notebook 07 nor the current neighborhood here constructs an interacting four-dimensional Yang–Mills measure.

References

[1] Pinned Yang–Mills-LLM-experiment account, commit `e392ef45013a581664bab76b908671 70b30fe10e`. The source passages used here are Raw I:13583–13593 (coarse penalty), 13638–13655 (extension normalization), 13698–13731 and 13811–13821 (current vertices and auxiliary inverse), 13774 (gauge warning), 23909–23944 (scalar average and projection), and 24349–24373 (corrected one-form average).

[2] T. Bałaban, *Renormalization group approach to lattice gauge field theories I*, Commun. Math. Phys. 109 (1987), 249–301. Copy of the original article.